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CS2403 COMPUTER NETWORKS

Mini Project Report

Secure Emergency Response Network

Submitted

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CERTIFICATE

Certified that the CS2403 Computer Networks Mini Project work titled *Secure Emergency Response Network* is carried out by Dhulipalla ShreeVathsa (1RUA25CSE0159), DIGANTH.R(1RUA25CSE0160), AYUSH GOWDA MJ (1RUA25CSE097), DISHANATH VASUPAL(1RUA25CSE0162), who are bonafide students of the School of Computer Science and Engineering, RV University, Bengaluru, during the year 2026–27. It is certified that all corrections/ suggestions from all the continuous internal evaluations have been incorporated into the project and in this report.

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Faculty Guide

Program Director

1 Problem statement

In an emergency such as a fire, a road accident or a natural disaster, the Emergency Operations Center (EOC), the hospital, the police and the fire department all have to pass information to each other quickly. If they are not connected, messages are delayed. If they are connected on one flat, open network, any department can reach the systems of every other department, including the administrative systems of the EOC. This project simulates a small network in Cisco Packet Tracer that connects these four departments through routers so that they can communicate, and uses Extended Access Control Lists (ACLs) to block selected traffic between departments. The project shows how IP addressing, OSPF routing, DHCP, VLANs and ACLs can be combined in one small emergency response network.

2 Introduction

The network connects four departments: the Emergency Operations Center (EOC), the Hospital, the Police Department and the Fire Department. Each department has its own router (Cisco 2911) and its own LAN built on a Cisco 2960 switch. The four routers are connected in a line: R1-EOC to R2-HOSP to R3-POLICE to R4-FIRE. The EOC has a Control PC, an Admin PC and the EOC Server. The Hospital has two PCs, the Police Department has three PCs and the Fire Department has three PCs.

Emergency departments must be able to talk to each other, but they should not all have access to everything. For example, the Admin network of the EOC is treated as the most sensitive part of the network, so the other departments are blocked from reaching it. This project tries to get both things in one design: connectivity between departments, and restrictions on who can reach what.

Every LAN has its own /24 subnet (192.168.10.0 to 192.168.40.0), and the routers are joined by small /30 links (10.0.12.0, 10.0.23.0 and 10.0.34.0). The routers use OSPF in area 0, so each router learns the networks of the other departments automatically and no static routes are needed.

Two more features are used at the EOC. DHCP is used on R1 so that the Control and Admin PCs get their addresses automatically; the PCs in the Hospital, Police and Fire departments have static IP addresses. VLANs are used to split one switch into three separate networks (Control, Admin and Server), and the router connects them using 802.1Q subinterfaces.

Security is added with Extended ACLs on all four routers. An ACL is a list of permit and deny rules that a router checks for every packet. Basic connectivity is tested with ping. In this report, the

configuration that was written is kept separate from the results that were actually verified, and places where a screenshot or result still has to be added are marked in square brackets.

3 Network Diagram

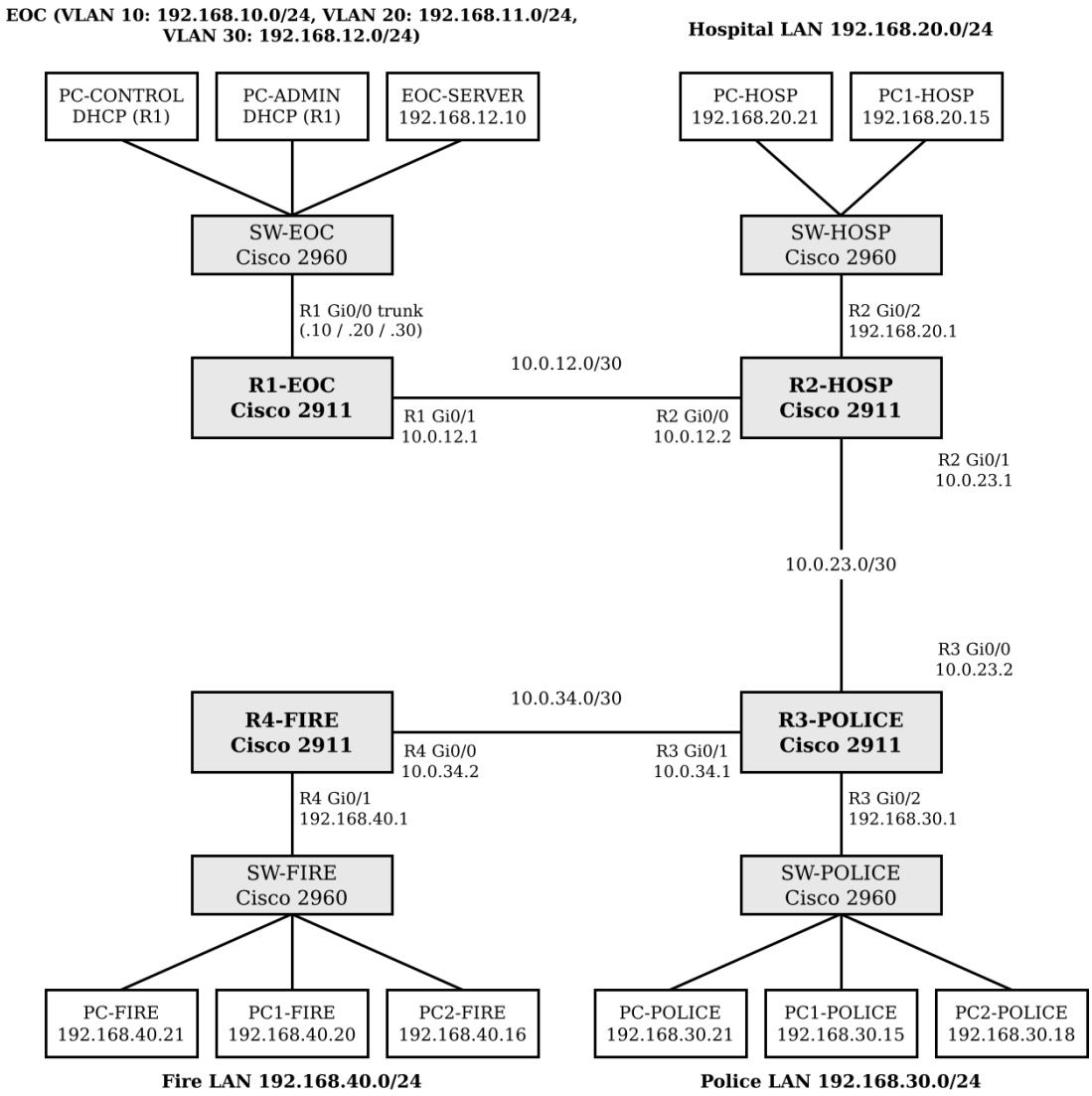


Figure 3.1: Network diagram of the Secure Emergency Response Network (drawn from the Packet Tracer project file; the routers form the chain R1-EOC ↔ R2-HOSP ↔ R3-POLICE ↔ R4-FIRE)

The routers are shown in a square only to fit the page. Logically they are connected in a line, R1 to R2 to R3 to R4, so traffic from the EOC to the Fire Department passes through the Hospital and Police routers. Tables 3.1 to 3.3 list the addresses used in the network.

Table 3.1: LAN subnets and default gateways

LAN	VLAN	Network	Default gateway	Router interface
EOC Control	10	192.168.10.0/24	192.168.10.1	R1 Gi0/0.10
EOC Admin	20	192.168.11.0/24	192.168.11.1	R1 Gi0/0.20
EOC Server	30	192.168.12.0/24	192.168.12.1	R1 Gi0/0.30
Hospital	–	192.168.20.0/24	192.168.20.1	R2 Gi0/2
Police	–	192.168.30.0/24	192.168.30.1	R3 Gi0/2
Fire	–	192.168.40.0/24	192.168.40.1	R4 Gi0/1

Table 3.2: Point-to-point links between routers

Link	Network	First router interface	Second router interface
EOC – Hospital	10.0.12.0/30	R1 Gi0/1: 10.0.12.1	R2 Gi0/0: 10.0.12.2
Hospital – Police	10.0.23.0/30	R2 Gi0/1: 10.0.23.1	R3 Gi0/0: 10.0.23.2
Police – Fire	10.0.34.0/30	R3 Gi0/1: 10.0.34.1	R4 Gi0/0: 10.0.34.2

Table 3.3: End devices (addresses as saved in the Packet Tracer file)

Device	Switch port	IP address	Gateway	How the address is assigned
PC-CONTROL	SW-EOC Fa0/1 (VLAN 10)	Not saved in file	192.168.10.1	DHCP (from R1)
PC-ADMIN	SW-EOC Fa0/2 (VLAN 20)	Not saved in file	192.168.11.1	DHCP (from R1)
EOC-SERVER	SW-EOC Fa0/3 (VLAN 30)	192.168.12.10	192.168.12.1	Static
PC-HOSP	SW-HOSP Fa0/1	192.168.20.21	192.168.20.1	Static
PC1-HOSP	SW-HOSP Fa0/3	192.168.20.15	192.168.20.1	Static
PC-POLICE	SW-POLICE Fa0/1	192.168.30.21	192.168.30.1	Static
PC1-POLICE	SW-POLICE Fa0/2	192.168.30.15	192.168.30.1	Static
PC2-POLICE	SW-POLICE Fa0/3	192.168.30.18	192.168.30.1	Static
PC-FIRE	SW-FIRE Fa0/1	192.168.40.21	192.168.40.1	Static
PC1-FIRE	SW-FIRE Fa0/2	192.168.40.20	192.168.40.1	Static
PC2-FIRE	SW-FIRE Fa0/3	192.168.40.16	192.168.40.1	Static

Note: DHCP is used only for the Control and Admin PCs at the EOC, so no IP address is saved for them in the project file. All other PCs and the EOC server have static addresses that were assigned by hand.

4 Configuration setup

This section first explains the technologies used and why each one is needed in this project, and then lists the configuration of every device. The configuration shown is taken from the saved Packet Tracer project file; blank lines, default settings and unused interfaces are left out to keep it short.

4.1 Technologies used

OSPF routing

OSPF (Open Shortest Path First) is a routing protocol in which routers tell each other which networks they are connected to. In this project all four routers run OSPF process 1 in area 0 and advertise their LANs and their point-to-point links. Because of this, the EOC router learns the Fire network and the Fire router learns the EOC networks without any static routes. If a new LAN is added, only one network statement has to be added.

IP addressing and subnetting

Every LAN has its own /24 network, so each department is a separate broadcast domain with its own gateway. The links between routers use /30 networks, which have only two usable addresses, so no addresses are wasted on a link that connects just two routers. The address plan is in Tables 3.1 and 3.2. Keeping each department in its own subnet is also what makes it possible to write ACL rules per department.

DHCP

DHCP gives IP address, gateway and DNS server to a PC automatically. In this project DHCP is used for the EOC Control and Admin PCs only. R1 is the DHCP server and has two pools, EOC-CONTROL and EOC-ADMIN. The first 20 addresses of each of these LANs are excluded so that they can be used for fixed devices, and the pools give 192.168.12.10 (the EOC server) as the DNS server. The PCs in the Hospital, Police and Fire departments are given static addresses by hand. The configuration of R2, R3 and R4 also contains DHCP pools, but they are not used because those PCs do not take addresses from DHCP.

VLAN segmentation

A VLAN splits one physical switch into separate logical networks. At the EOC, a single switch (SW-EOC) carries three VLANs: VLAN 10 for Control, VLAN 20 for Admin and VLAN 30 for the Server. The switch port towards R1 is a trunk that allows VLANs 10, 20 and 30. On R1, one physical interface (Gi0/0) is divided into three subinterfaces (Gi0/0.10, .20, .30) using 802.1Q encapsulation, and each subinterface is the gateway of one VLAN. This method is called router-on-a-stick. VLANs

are used here so that the Admin PC and Control PC are on different subnets even though they are on one switch, which makes it possible to block traffic between them. VLANs are used only at the EOC; the other departments use a single default LAN.

Extended ACLs

A standard ACL checks only the source address, but an extended ACL can check source, destination and protocol (for example only ping). This is needed here because the rules are of the type "this department must not reach that department". Each router has one extended ACL applied inbound on its LAN interface, so unwanted packets are dropped as soon as they enter the router. The policies are explained in Section 4.3.

Router and switch configuration

The routers are configured with interface addresses, OSPF, DHCP pools and ACLs. The switches are configured with access ports and, at the EOC, a trunk port. The PCs and the server are configured with an IP address, subnet mask and default gateway, either manually or by DHCP.

Network connectivity testing

Ping (ICMP echo) is used to check whether one device can reach another. Commands such as show ip route, show ip ospf neighbor, show ip dhcp binding and show access-lists are used to check routing, DHCP and the ACL hit counters. The planned tests are in Section 5.

4.2 Device configuration

The end devices are listed first, then the switches and then the routers.

PC-CONTROL –

Connected to SW-EOC Fa0/1 (VLAN 10). Default gateway 192.168.10.1. It gets its address by DHCP from the EOC-CONTROL pool on R1 (192.168.10.21 onwards), so no IP address is saved in the file.

PC-ADMIN –

Connected to SW-EOC Fa0/2 (VLAN 20). Default gateway 192.168.11.1. It gets its address by DHCP from the EOC-ADMIN pool on R1 (192.168.11.21 onwards), so no IP address is saved in the file.

EOC-SERVER –

Connected to SW-EOC Fa0/3 (VLAN 30). IP address 192.168.12.10, subnet mask 255.255.255.0, default gateway 192.168.12.1. This address is given as the DNS server in the DHCP pools. Whether a DNS service is configured on the server is [not verified].

PC-HOSP and PC1-HOSP –

Connected to SW-HOSP Fa0/1 and Fa0/3. PC-HOSP has 192.168.20.21 and PC1-HOSP has 192.168.20.15 (both static). Subnet mask 255.255.255.0, gateway 192.168.20.1.

PC-POLICE, PC1-POLICE and PC2-POLICE –

Connected to SW-POLICE Fa0/1, Fa0/2 and Fa0/3. Static addresses 192.168.30.21, 192.168.30.15 and 192.168.30.18. Subnet mask 255.255.255.0, gateway 192.168.30.1.

PC-FIRE, PC1-FIRE and PC2-FIRE –

Connected to SW-FIRE Fa0/1, Fa0/2 and Fa0/3. Static addresses 192.168.40.21, 192.168.40.20 and 192.168.40.16. Subnet mask 255.255.255.0, gateway 192.168.40.1.

SW-EOC (Cisco 2960) –

```
hostname SW-EOC
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
interface FastEthernet0/2
 switchport access vlan 20
 switchport mode access
 spanning-tree portfast
interface FastEthernet0/3
 switchport access vlan 30
 switchport mode access
 spanning-tree portfast
interface GigabitEthernet0/1
 switchport trunk allowed vlan 10,20,30
 switchport mode trunk
```

SW-HOSP (Cisco 2960) –

```
hostname Switch
interface FastEthernet0/2
 switchport access vlan 20
 switchport mode access
```

SW-HOSP, SW-POLICE and SW-FIRE keep the default hostname “Switch” and all their ports are in the default VLAN 1, so each is a plain switch with no VLAN or trunk configuration. The only exception is Fa0/2 on SW-HOSP, which is set to VLAN 20, but no device is connected to that port.

SW-POLICE and SW-FIRE (Cisco 2960) –

No configuration other than the default (all ports in VLAN 1).

R1-EOC (Cisco 2911) –

```
hostname R1-EOC
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp excluded-address 192.168.11.1 192.168.11.20
ip dhcp pool EOC-CONTROL
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 192.168.12.10
ip dhcp pool EOC-ADMIN
 network 192.168.11.0 255.255.255.0
 default-router 192.168.11.1
 dns-server 192.168.12.10
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip access-group ADMIN-PROTECTION in
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.11.1 255.255.255.0
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.12.1 255.255.255.0
interface GigabitEthernet0/1
 ip address 10.0.12.1 255.255.255.252
router ospf 1
 log-adjacency-changes
 network 10.0.12.0 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.11.0 0.0.0.255 area 0
 network 192.168.12.0 0.0.0.255 area 0
ip access-list extended ADMIN-PROTECTION
 deny ip 192.168.10.0 0.0.0.255 192.168.11.0 0.0.0.255
 permit ip any any
```

R2-HOSP (Cisco 2911) –

```
hostname R2-HOSP
ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp pool HOSPITAL
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 192.168.12.10
interface GigabitEthernet0/0
 ip address 10.0.12.2 255.255.255.252
interface GigabitEthernet0/1
 ip address 10.0.23.1 255.255.255.252
interface GigabitEthernet0/2
 ip address 192.168.20.1 255.255.255.0
 ip access-group HOSPITAL-ACL in
router ospf 1
 log-adjacency-changes
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.23.0 0.0.0.3 area 0
 network 192.168.20.0 0.0.0.255 area 0
ip access-list extended HOSPITAL-ACL
 deny ip 192.168.20.0 0.0.0.255 192.168.11.0 0.0.0.255
 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 deny icmp 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255 echo
 permit icmp 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255 echo-reply
 permit ip 192.168.20.0 0.0.0.255 any
```

R3-POLICE (Cisco 2911) –

```
hostname R3-POLICE
ip dhcp excluded-address 192.168.30.1 192.168.30.20
ip dhcp pool POLICE
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 192.168.12.10
interface GigabitEthernet0/0
 ip address 10.0.23.2 255.255.255.252
interface GigabitEthernet0/1
 ip address 10.0.34.1 255.255.255.252
interface GigabitEthernet0/2
 ip address 192.168.30.1 255.255.255.0
 ip access-group SERVER-PROTECTION in
router ospf 1
 log-adjacency-changes
 network 10.0.23.0 0.0.0.3 area 0
 network 10.0.34.0 0.0.0.3 area 0
 network 192.168.30.0 0.0.0.255 area 0
ip access-list extended SERVER-PROTECTION
 deny ip 192.168.30.0 0.0.0.255 192.168.11.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 permit ip 192.168.30.0 0.0.0.255 any
```

R4-FIRE (Cisco 2911) –

```
hostname R4-FIRE
ip dhcp excluded-address 192.168.40.1 192.168.40.20
ip dhcp pool FIRE
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
 dns-server 192.168.12.10
interface GigabitEthernet0/0
 ip address 10.0.34.2 255.255.255.252
interface GigabitEthernet0/1
 ip address 192.168.40.1 255.255.255.0
 ip access-group FIRE-PROTECTION in
router ospf 1
 log-adjacency-changes
 network 10.0.34.0 0.0.0.3 area 0
 network 192.168.40.0 0.0.0.255 area 0
ip access-list extended FIRE-PROTECTION
 deny ip 192.168.40.0 0.0.0.255 192.168.11.0 0.0.0.255
 permit ip 192.168.40.0 0.0.0.255 any
```

On R1 the physical interface Gi0/0 has no IP address of its own; it only carries the three dot1Q subinterfaces. All routers use OSPF process 1 in area 0.

4.3 Security policies (ACLs)

Four extended ACLs are configured, one on each router, and each is applied inbound on the LAN interface of that router. The table below describes what each ACL does according to the saved configuration. It describes the intended behaviour; whether each one works as intended is checked in Section 5.

Table 4.1: Extended ACLs in the network

ACL name	Router	Applied on	What the rules do
ADMIN-PROTECTION	R1	Gi0/0.10 (in)	Denies all IP traffic from the Control subnet (192.168.10.0/24) to the Admin subnet (192.168.11.0/24). Everything else is permitted.
HOSPITAL-ACL	R2	Gi0/2 (in)	Denies all IP traffic from the Hospital subnet to the Admin subnet and to the Police subnet. Denies ping requests (ICMP echo) from the Hospital to the Fire subnet, but permits ping replies (echo-reply) to Fire. Everything else is permitted.
SERVER-PROTECTION	R3	Gi0/2 (in)	Denies all IP traffic from the Police subnet to the Admin subnet and to the Hospital subnet. Everything else from Police is permitted.
FIRE-PROTECTION	R4	Gi0/1 (in)	Denies all IP traffic from the Fire subnet to the Admin subnet. Everything else is permitted.

Points to remember about these ACLs

- Direction and placement matter. An inbound ACL only checks packets that enter that interface. A ping needs a request and a reply, and the reply is also checked by whichever ACL it passes. For example, ADMIN-PROTECTION blocks Control to Admin, but it does not stop a ping started from Admin to Control. The request reaches the Control PC, but the reply travels from Control to Admin, and that is denied. So the ping fails in both directions, even though only one direction is written in the ACL.
- The HOSPITAL-ACL line that denies echo to the Fire subnet and the line that permits echo-reply are written so that the Fire Department can ping the Hospital and get a reply, while the Hospital cannot start a ping to the Fire Department.
- The ACL named SERVER-PROTECTION is on R3 and restricts traffic from the Police subnet to the Admin and Hospital subnets. It does not specifically block the Police subnet from the EOC server subnet (192.168.12.0/24), so the name is a little misleading.
- The HOSPITAL-ACL blocks all IP traffic from Hospital to Admin and Police, but towards the Fire subnet it blocks only ping requests. Other traffic from the Hospital to the Fire subnet is permitted by the last line.
- ACL rules are checked from top to bottom and the first match is used. Any packet that matches no line is dropped because of the implicit deny at the end.

5 Results

This section lists what should happen when the network is tested, based on reading the configuration, and has space for the proof (screenshots) that has to be added from Packet Tracer. The “expected” column is my reading of the ACLs and routing; it is not a test result. No test result has been entered where it has not been verified.

5.1 Routing checks

Table 5.1: Routing and neighbour checks

Check	Expected from configuration
show ip ospf neighbor on R1	R1 sees R2 as a neighbour on Gi0/1
show ip ospf neighbor on R2	R2 sees R1 (Gi0/0) and R3 (Gi0/1)
show ip ospf neighbor on R3	R3 sees R2 (Gi0/0) and R4 (Gi0/1)
show ip route on R1	OSPF routes (O) to 192.168.20.0, 192.168.30.0, 192.168.40.0, 10.0.23.0 and 10.0.34.0
show ip route on R4	OSPF routes (O) to 192.168.10.0, .11.0, .12.0, .20.0 and .30.0

5.2 Connectivity and ACL tests

Table 5.2: Planned ping tests and what the ACLs should do

No.	From → To	ACL effect	Expected from configuration
1	PC-CONTROL → EOC-SERVER	Permitted by ADMIN-PROTECTION	Reply received
2	PC-CONTROL → PC-ADMIN	Denied by ADMIN-PROTECTION	Blocked
3	PC-ADMIN → PC-CONTROL	Request not filtered, but the reply is denied by ADMIN-PROTECTION	No reply (return traffic blocked)
4	PC-ADMIN → EOC-SERVER	No ACL on VLAN 20 or VLAN 30 interfaces	Reply received
5	PC-HOSP → EOC-SERVER	Permitted by HOSPITAL-ACL	Reply received
6	PC-HOSP → PC-CONTROL	Permitted by HOSPITAL-ACL; reply permitted by ADMIN-PROTECTION	Reply received
7	PC-HOSP → PC-ADMIN	Denied by HOSPITAL-ACL	Blocked
8	PC-HOSP → PC-POLICE	Denied by HOSPITAL-ACL	Blocked
9	PC-HOSP → PC-FIRE	Ping request denied by HOSPITAL-ACL	Blocked
10	PC-FIRE → PC-HOSP	Request permitted by FIRE-PROTECTION; echo-reply permitted by HOSPITAL-ACL	Reply received
11	PC-POLICE → EOC-SERVER	Permitted by SERVER-PROTECTION	Reply received
12	PC-POLICE → PC-ADMIN	Denied by SERVER-PROTECTION	Blocked
13	PC-POLICE → PC-HOSP	Denied by SERVER-PROTECTION	Blocked
14	PC-POLICE → PC-FIRE	Permitted by SERVER-PROTECTION and FIRE-PROTECTION	Reply received
15	PC-FIRE → PC-POLICE	Permitted by FIRE-PROTECTION and SERVER-PROTECTION	Reply received
16	PC-FIRE → PC-ADMIN	Denied by FIRE-PROTECTION	Blocked
17	PC-FIRE → EOC-SERVER	Permitted by FIRE-PROTECTION	Reply received

A “Blocked” result means the ping does not succeed. Packet Tracer may show it as “Destination host unreachable” or as a request time out; the exact message should be copied from the screenshot. The ACL hit counters from show access-lists on each router can be used as extra proof that the right rule was matched.

5.3 DHCP check

DHCP is used only for the EOC Control and Admin PCs, and R1 is the DHCP server for both. The Hospital, Police and Fire PCs have static addresses, so they are not part of this check. The result has to be verified and entered below.

Table 5.3: DHCP verification (R1)

Pool on R1	Addresses given out
EOC-CONTROL	192.168.10.21 onwards
EOC-ADMIN	192.168.11.21 onwards

The ACL on the Control interface (ADMIN-PROTECTION) ends with permit ip any any, and the Admin interface has no ACL, so the ACLs are not expected to stop DHCP requests at the EOC. The ipconfig output of PC-CONTROL and PC-ADMIN should be added as proof.

5.4 Proof of communication

At least three clear screenshots from Packet Tracer should be pasted below.

```
R1-EOC>show ip route ospf
     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O       10.0.23.0 [110/2] via 10.0.12.2, 00:08:50, GigabitEthernet0/1
O       10.0.34.0 [110/3] via 10.0.12.2, 00:08:50, GigabitEthernet0/1
O    192.168.20.0 [110/2] via 10.0.12.2, 00:09:00, GigabitEthernet0/1
O    192.168.30.0 [110/3] via 10.0.12.2, 00:08:50, GigabitEthernet0/1
O    192.168.40.0 [110/4] via 10.0.12.2, 00:08:50, GigabitEthernet0/1

R1-EOC>
```

Figure 5.1: Output of show ip route on R1

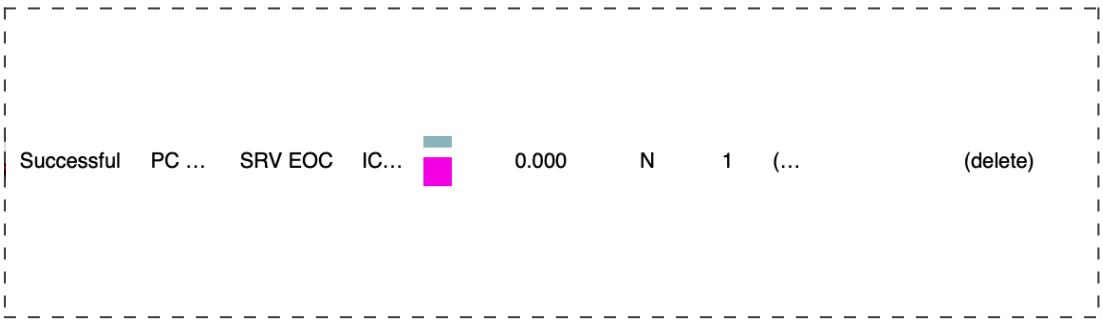


Figure 5.2: Ping from PC-CONTROL to EOC-SERVER

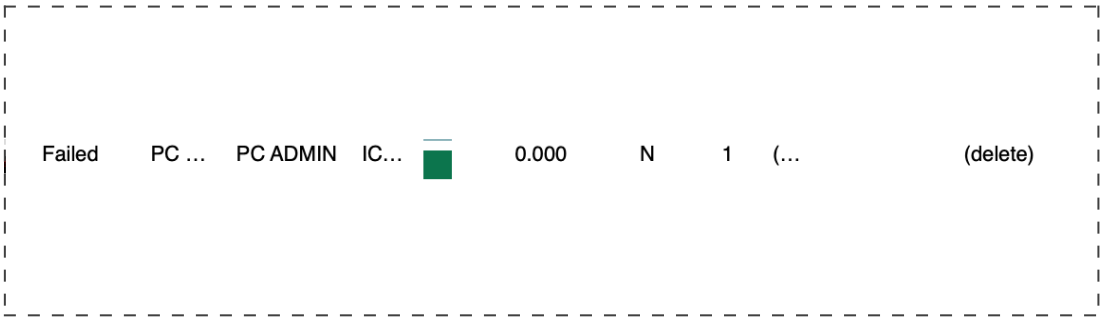


Figure 5.3: Ping from PC-CONTROL to PC-ADMIN (blocked case)

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.12.1      1   FULL/BDR        00:00:36    10.0.12.1       GigabitEthernet0/0
192.168.30.1      1   FULL/DR         00:00:36    10.0.23.2       GigabitEthernet0/1
R2-HOSP>
```

Figure 5.4: Output of show ip ospf neighbor on R2

```
R1-EOC>show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.21    0002.4AEC.A219           --                     Automatic
192.168.11.21    0001.C7B8.B5EB           --                     Automatic
R1-EOC>
```

Figure 5.5: DHCP leases on R1

```
R2-HOSP>enable
R2-HOSP#show access-lists
Extended IP access list HOSPITAL-ACL
    10 deny ip 192.168.20.0 0.0.0.255 192.168.11.0 0.0.0.255
    20 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
    30 deny icmp 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255 echo
    40 permit icmp 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255 echo-reply
    50 permit ip 192.168.20.0 0.0.0.255 any

R2-HOSP#
```

Figure 5.6: Output of show access-lists on R2

5.5 Limitations

- DHCP is used only at the EOC. The other departments use static addresses, which is simple but means every new device has to be configured by hand.
- VLANs are used only at the EOC. The Hospital, Police and Fire LANs each use one default VLAN, so VLAN segmentation is not applied inside those departments.

- The network is a straight line, so if one link or router fails the departments beyond it lose connection to the others. There is no backup path.
- No passwords are set on the console or vty lines (the vty lines only have the login command) and OSPF has no authentication, so the devices themselves are not secured.
- The ACLs are based on source and destination subnets only. They do not look at users or applications, and the exact behaviour depends on where each ACL is applied, as explained in Section 4.3.
- The DNS address given by DHCP is that of the EOC server, but a DNS service on the server has not been verified.

5.6 Future enhancements

- Use DHCP in the Hospital, Police and Fire departments as well. The ACLs there would need a rule that permits DHCP requests (UDP port 68 to 67) before they are tested.
- Add a second link between R1 and R4 to form a ring so that the network still works if one link fails.
- Set passwords or SSH on the devices and use OSPF authentication.
- Use VLANs in the other departments as well, and use ACL logging to see which packets are being blocked.
- Use one DHCP server at the EOC with ip helper-address on the other routers, so that all addresses are managed in one place.

6 Conclusion

In this project a network for four emergency departments (EOC, Hospital, Police and Fire) was built in Cisco Packet Tracer using four Cisco 2911 routers and four Cisco 2960 switches. Each department was given its own subnet, the routers were connected with /30 links and OSPF area 0 was used so that all networks are learned automatically. DHCP was configured on R1 for the Control and Admin PCs, the other PCs were given static addresses, VLANs and a trunk were used at the EOC, and an Extended ACL was applied on each router to restrict traffic towards the Admin subnet and between some departments. Such a design is useful because departments can exchange information while sensitive networks are protected from the others. The expected behaviour of the ACLs and the routing is listed in Section 5, but the ping results and screenshots still have to be added. In future the network can be extended with a backup link, secured device access, a central DHCP server and VLANs in every department.